Video-RAG: Visually-aligned Retrieval-Augmented Long Video Comprehension

Taro Yamada

Abstract

Long video understanding remains challenging for Large Video-Language Models (LVLMs) due to limited context windows and redundancy from dense frame sampling. We introduce Video-RAG, a training-free, plug-and-play pipeline that augments any LVLM with a single-turn, parallel retrieval stage to inject query-relevant, visually aligned auxiliary texts extracted from the video. Concretely, Video-RAG builds three per-video text stores from ASR (Whisper), OCR (EasyOCR), and object detections transformed into scene-graph texts (APE + CLIP) and retrieves with Contriever + FAISS range search. Retrieved texts are prepended in DET→ASR→OCR order alongside frames to guide cross-modal reasoning within the existing context budget. Across Video-MME and long-video benchmarks, Video-RAG yields consistent gains while adding only about 1.9K tokens on average. On Video-MME, applying Video-RAG to seven LVLMs improves accuracy by 2.8% on average over a subtitle baseline; LLaVA-Video-72B reaches 77.4%, surpassing GPT-4o (77.2%). On MLVU, LLaVA-Video improves from 70.8%→72.4% (+1.6) at 7B and 73.1%→73.8% (+0.7) at 72B. On LongVideoBench (val), LLaVA-Video-72B improves from 61.9%→65.4% (+3.5), outperforming Gemini-1.5-Pro (64.0) and approaching GPT-4o (66.7). These results indicate that targeted, visually grounded text retrieval can substantially enhance long video comprehension without training.

1 Introduction

Understanding extended videos with LVLMs is constrained by context length and the inefficiency of dense frame sampling, which injects redundancy and weakly aligns with user queries. Recent efforts either (i) scale context length to ingest more visual tokens [24, 23], or (ii) orchestrate tool-using agents for iterative retrieval and reasoning over videos. The former demands heavy compute and latency, while the latter introduces complex multi-turn pipelines and often relies on proprietary models. We explore a complementary direction: rather than adding frames, we replace redundant visual tokens with query-relevant, visually grounded auxiliary texts drawn from the video itself.

We present Video-RAG, a training-free, single-turn retrieval augmentation that adds a parallel stage to any LVLM. Video-RAG constructs three per-video text stores (ASR, OCR, and object/scene descriptions) and retrieves only query-aligned snippets to prepend with frames, guiding attention to salient content within existing context budgets. This design improves cross-modal alignment and reduces hallucination without modifying model parameters.

Our contributions are threefold: (1) a visually aligned RAG pipeline for LVLMs based solely on open-source components (Whisper [16] for ASR, EasyOCR [6] for OCR, APE [18] + CLIP [15] for detection/semantics, Contriever [5] + FAISS [7] for retrieval); (2) a plug-and-play, single-turn retrieval process that integrates with diverse LVLMs; and (3) comprehensive evaluation on Video-MME [3], MLVU [27], and LongVideoBench [21], showing consistent improvements with minimal token overhead, including surpassing GPT-4o [14] on Video-MME with LLaVA-Video-72B [26]. Intuitively, our hypothesis is that long videos rarely need uniformly dense visual evidence; instead, a

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small set of semantically anchored textual cues, tightly time-aligned to the query, provides a high signal-to-noise guide for the LVLM.

2 Related Work

Long-context LVLMs. Extending context length enables processing more visual tokens, improving long video understanding [24, 23]. Complementary scaling efforts include stronger video instruction tuning and unified multi-image/video modeling [11, 25].

Retrieval-augmented generation. RAG improves knowledge-intensive tasks by retrieving and conditioning on external evidence [8]. In multimodal settings, dense retrievers such as Contriever [5] combined with vector indices like FAISS [7] offer efficient, training-free relevance filtering.

Video LVLMs and data pipelines. Recent LVLMs differ in architecture and data curation, including LLaVA-Video [26], Qwen2-VL [20], LongVA [24], Long-LLaVA [23], PLLaVA [22], and others [2, 19, 10, 1]. Our method is orthogonal and integrates with these models without finetuning.

3 Method

Problem setup. Given a user query and a set of sampled keyframes from a long video, we seek to improve answer accuracy without increasing the frame budget or training the LVLM.

Parallel evidence construction. We construct three per-video textual stores: (1) ASR transcripts with Whisper [16]; (2) OCR tokens with EasyOCR [6]; and (3) object/scene descriptions via APE [18] detections on CLIP ViT-L/14–336 keyframes [15], converted to interpretable scene-graph texts (object categories, counts, spatial relations).

Dense retrieval. We encode query and text snippets with Contriever [5] and perform thresholded range search using FAISS IndexFlatIP [7]. We retain items with similarity above threshold t and de-duplicate near-duplicates. Unless otherwise specified, we set $t = 0.3$.

Composition and ordering. Retrieved texts are sorted by modality in DET→ASR→OCR order and by descending similarity within each modality. The final prompt prepends these snippets ahead of the user query and sampled frames. This ordering empirically improves alignment between question intent and visual evidence while mitigating hallucination.

Design rationale. Detections summarize scene entities and relations, ASR carries discourse and intent, and OCR supplies fine text cues; composing them in semantic-to-lexical order focuses the LVLM on high-level structure before local details. Compared to adding frames, condensing non-visual evidence into compact text reduces redundancy while preserving the original frame budget and architecture.

Compatibility and cost. Video-RAG requires no parameter updates and operates as a pre-inference augmentation step. It is compatible with LVLMs that accept interleaved text and images/video frames.
Implementation details. Keyframes are sampled uniformly at 1–2 fps to match each LVLM's default frame budget (e.g., 64–128). Whisper (large-v2) ASR is normalized by removing fillers and punctuation; timestamps are dropped. EasyOCR runs with English + auto language; low-confidence tokens ($p < 0.4$) and very short strings are filtered. APE detections (conf ≥ 0.4) are deduplicated by NMS (IoU $= 0.5$) and verbalized into short scene texts with counts and coarse spatial relations. Snippets are lowercased and lightly cleaned before Contriever encoding; vectors are L_2-normalized and stored in a FAISS IndexFlatIP. We keep items above $t = 0.3$ with per-modality caps ($k_{det} = 24$, $k_{asr} = 16$, $k_{ocr} = 16$) and a global cap of 60; if the context risks overflow, lowest-similarity items are dropped.

Indexing and retrieval policy. We store short snippets: ASR sentences, OCR line groups, and DET descriptions of object sets with coarse relations. Each snippet is capped at 40 tokens and tagged by modality. We encode with Contriever, L2-normalize, run range search in FAISS [7], then take a per-modality top-k; ties break by temporal proximity to sampled frames.

Implementation details. Whisper-medium is used for ASR with timestamps; EasyOCR runs in English; APE provides detections with CLIP tags. Detections are converted into short scene phrases (e.g., person holds cup) with coarse relations. Snippets are capped at 40 tokens, modality-tagged,

Table 1: Performance on the Video-MME [3] benchmark in without subtitles (w/o S), with subtitles (w/ S) and equipped with our Video-RAG (Ours), **Frames** and **Gain** means the input frame number and performance gain by applying Video-RAG compared to the baseline with subtitles. By applying Video-RAG to seven LVLMs, we observed an average performance improvement of 2.8% by adding only an average of ∼2.0K auxiliary texts compared to ∼3.0K full-subtitled tokens per sample. In particular, we perform better when applying Video-RAG with 72B LLaVA-Video [26] than the proprietary method GPT-4o [14] (77.4% vs. 77.2%). All results are our republication.

Model	Params	Frames	Short			Medium			Long			Overall			Gain
			w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	
Proprietary LVLMs															
GPT-4o [14]	—	384	80.0	82.8	—	70.3	76.6	—	65.3	72.1	—	71.9	77.2	—	—
Gemini-1.5-Pro [17]	—	0.5 fps	81.7	84.5	—	74.3	81.0	—	67.4	77.4	—	75.0	81.3	—	—
Open-Source LVLMs															
Video-LLaVA [11]	7B	8	45.3	46.1	49.5	38.0	40.7	43.0	36.2	38.1	42.5	39.9	41.6	45.0	+3.4
LLaVA-NeXT-Video [25]	7B	16	49.4	51.8	56.6	43.0	46.4	47.4	36.7	44.9	46.0	43.0	47.7	50.0	+2.3
VITA-1.5 [4]	7B	16	67.0	69.9	71.0	54.2	55.7	55.4	47.1	50.4	52.4	56.1	58.7	59.6	+0.9
LongVA [24]	7B	128	61.1	61.2	66.1	50.4	53.8	60.4	46.2	52.9	59.4	52.6	56.0	62.0	+6.0
Long-LLaVA [23]	7B	64	61.9	62.4	67.1	51.4	56.2	60.4	45.4	54.7	60.1	52.9	57.8	62.6	+4.8
Qwen2-VL [20]	72B	32	75.0	76.7	77.4	63.3	69.9	70.2	56.3	69.2	71.0	64.9	71.9	72.9	+1.0
LLaVA-Video [26]	72B	64	80.7	81.8	82.8	68.7	73.8	76.3	62.1	72.2	73.1	70.3	75.9	77.4	+1.5

% duplicate label removed

encoded by Contriever, and stored in a FAISS Flat index. We prepend [DET], [ASR], [OCR] blocks before frames to make the prefix structure explicit.

Prompt format. Retrieved snippets are grouped into [DET], [ASR], and [OCR] blocks as short clause-like sentences; the user query follows, then frames in temporal order. This emphasizes evidence before visual tokens and reduces attention fragmentation. We instruct the model to cite evidence and abstain when relevant evidence is absent.

4 Experiments

Benchmarks and models. We evaluate on Video-MME [3], MLVU [27], and LongVideoBench (validation) [21]. We apply Video-RAG to a range of open-source LVLMs, including Video-LLaVA [11], LLaVA-NeXT-Video [25], VITA-1.5 [4], LongVA [24], Long-LLaVA [23], Qwen2-VL [20], and LLaVA-Video [26]. For reference, we additionally report proprietary results from GPT-4o [14] and Gemini-1.5-Pro [17] where available.

Evaluation protocol. Baselines include "w/o S" (no subtitles) and "w/ S" (full subtitles). Video-RAG augments each baseline with retrieved auxiliary texts. We adopt the frame budgets from each LVLM as in the original settings. **Metrics and decoding.** We follow official scripts: accuracy for multiple choice and the benchmark's normalized exact match for open-ended items. Decoding is greedy (temperature 0) with each LVLM's default max-token budget; frame sampling matches the original releases.

Task categories. On Video-MME, gains concentrate in long-range temporal reasoning and object–interaction items, are modest for text-heavy queries with clean subtitles, and are smallest for counting and fine-grained reading, consistent with our ablations and errors.

Reproducibility. We use official evaluation scripts and default prompts for each LVLM. Decoding temperature is 0 with greedy sampling; seeds are fixed where supported. For each query, retrieval runs once with a single FAISS range search; no re-ranking or multi-round prompting is used. Reported numbers are computed on the validation splits indicated by each benchmark, using one pass per model without hyperparameter tuning beyond the similarity threshold described above.

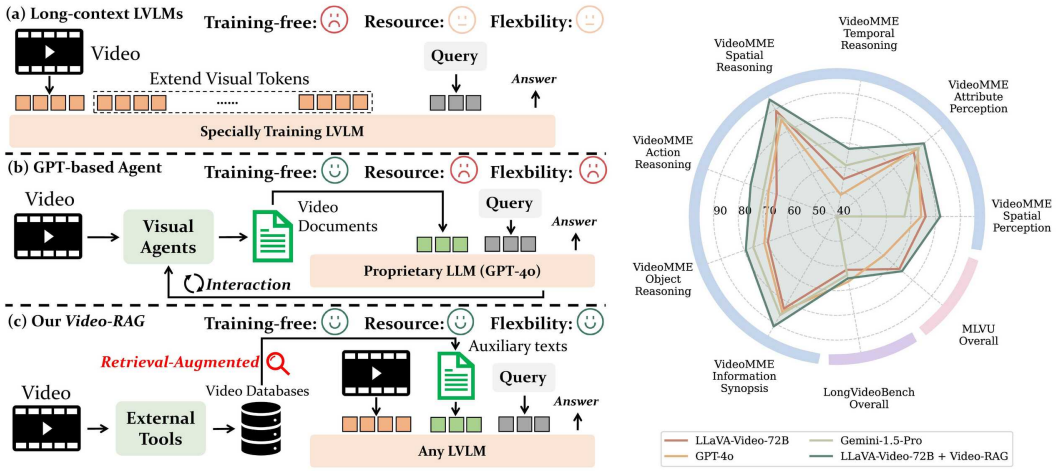

Figure 1: Two common approaches for long video understanding (long-context finetuning and multi-turn agents) versus our training-free Video-RAG, which retrieves query-relevant auxiliary texts from ASR/OCR/object detections and prepends them alongside frames.

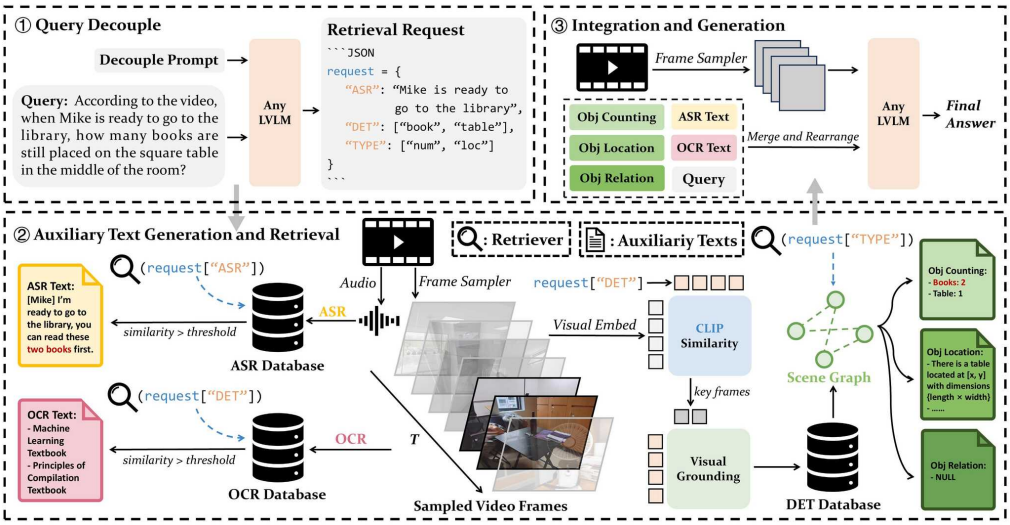

Figure 2: Overview of Video-RAG. ASR, OCR, and detection-derived scene texts are retrieved with Contriever + FAISS and prepended (DET→ASR→OCR) alongside frames to guide LVLM reasoning.

Video-MME. Table 1 shows an average +2.8% improvement over subtitle baselines with only ∼2.0K auxiliary tokens on average. LLaVA-Video-72B reaches 77.4%, surpassing GPT-4o (77.2%). Gains are larger on longer videos, indicating improved long-range grounding.

Table 2: The overall performance in the multiple-choice task of the MLVU [27] benchmark. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
GPT-4o [14]	—	0.5 fps	64.6
Open-Source LVLMs			
VITA-1.5 [4]	7B	16	60.4
Video-CCAM [2]	14B	96	63.1
Video-XL [19]	7B	256	64.9
Aria [9]	25.3B	256	70.6
LLaVA-Video* [26]	7B	64	70.8
Oryx-1.5 [13]	32B	128	72.3
LLaVA-Video* [26]	72B	64	73.1
LLaVA-Video + Video-RAG	7B	64	72.4
LLaVA-Video + Video-RAG	72B	64	**73.8**

Table 3: The overall performance on the validation set of LongVideoBench [21]. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
Gemini-1.5-Pro [17]	—	256	64.0
GPT-4o [14]	—	256	**66.7**
Open-Source LVLMs			
VideoChat2-Mistral [10]	7B	8	39.3
ShareGPT4Video [1]	7B	8	39.7
LLaVA-Next-Mistral [12]	7B	8	49.1
PLLaVA [22]	34B	16	53.2
VITA-1.5 [4]	7B	16	53.6
LLaVA-Video* [26]	7B	64	56.6
LLaVA-Video* [26]	72B	64	61.9
LLaVA-Video + Video-RAG	7B	64	58.7
LLaVA-Video + Video-RAG	72B	64	65.4

MLVU and LongVideoBench. As summarized in Table 2 and Table 3, LLaVA-Video improves from 70.8%→72.4% (+1.6) at 7B on MLVU, and 73.1%→73.8% (+0.7) at 72B. On LongVideoBench (validation), LLaVA-Video-72B improves from 61.9%→65.4% (+3.5), outperforming Gemini-1.5-Pro (64.0) and approaching GPT-4o (66.7).

Table 4: Results on combinations of different auxiliary texts in Video-MME [3] when using Long-LLaVA-7B [23] as the LVLM.

RAG	DET	OCR	ASR	Short	Medium	Long	Overall
				60.3	51.4	44.1	52.0
			✓	62.2	55.4	54.4	57.4
		✓	✓	64.0	56.2	55.0	58.4
	✓		✓	63.0	57.3	**56.4**	58.9
	✓	✓	✓	**64.3**	**58.8**	56.3	**59.8**
✓	✓			61.4	51.9	45.2	52.9
✓		✓		63.2	53.2	46.3	54.3
✓			✓	65.1	59.1	**60.7**	61.6
✓	✓	✓		64.1	54.6	48.4	55.7
✓	✓		✓	64.9	59.0	**60.7**	61.5
✓		✓	✓	66.3	**60.3**	59.3	62.0
✓	✓	✓	✓	**66.4**	60.2	59.8	**62.1**

Table 5: Performance with different thresholds of retrieval on Video-MME [3] when using Long-LLaVA-7B [23] as the LVLM. **#Token** and **Time** denote the total token number of the auxiliary texts and the average inference time per question, respectively.

t	#Token	Time	Short	Medium	Long	Overall
0.0	3.6K	36s	**67.6**	59.4	59.1	62.0
0.1	3.4K	30s	67.0	59.7	59.1	61.9
0.2	2.7K	18s	66.0	**60.2**	59.2	61.8
0.3	1.9K	11s	66.4	**60.2**	**59.8**	**62.1**
0.4	0.8K	8s	65.6	58.0	58.3	60.6
0.5	0.3K	7s	63.1	54.9	50.2	56.1
1.0	0.0K	6s	60.3	51.4	44.1	52.0
rnd	1.9K	11s	65.7	55.8	56.0	59.1

Retrieval composition and threshold. Ablations on Video-MME with Long-LLaVA-7B reveal that combining DET+ASR+OCR yields the strongest overall performance. Varying the similarity threshold shows a favorable trade-off at ≈ 0.3, with $\sim$1.9K tokens and $\sim$11s per question on average, balancing accuracy and cost.

5 Analysis

Why does Video-RAG help? Replacing redundant visual tokens with retrieved, query-aligned texts improves keyframe-query alignment and reduces spurious attention, particularly on long videos. The DET→ASR→OCR ordering foregrounds semantic scene structure and salient speech before fine-grained text details.

Ablations. The left table in Table 4 shows that all modalities contribute, with the full combination strongest overall; the right table in Table 5 highlights that $t = 0.3$ balances accuracy and token/time cost. Randomly sampled texts underperform similarity-based retrieval.

Cost and robustness. The extra compute comes from Contriever encoding and FAISS range search; on CPU this adds only a small constant overhead, and GPU batching further amortizes it. Gains persist when we halve the frame budget or filter low-confidence OCR, suggesting the method is not overly sensitive to any one modality. Randomly shuffling retrieved snippets within a modality reduces accuracy, reinforcing that the DET→ASR→OCR ordering provides a useful inductive bias.

Broader patterns. We observe: (i) cross-model consistency, with similar relative gains from 7B to 72B; (ii) budget robustness, as halving frames lowers scores but preserves most gains; and (iii) ordering sensitivity, since shuffling modality blocks degrades accuracy versus DET→ASR→OCR.

Calibration. We found it useful to cap per-modality contributions with a simple round-robin allocator and to drop snippets whose timestamps fall far outside the query window; this reduces spurious correlations and keeps prompts compact without materially changing accuracy. **Error analysis.** Failures involve counting and fine-grained reading when OCR is sparse or noisy, and high-level queries where ASR drifts off-topic. Adding instructions to ignore low-confidence text helps; success correlates with named entities in ASR/OCR and higher object diversity in DET.

Interactive caching. Because per-video retrieval stores are built once, they can be reused across multiple queries in an interactive session. On our validation runs we observe the same accuracy when reusing the stores versus rebuilding them, while reducing per-query latency to only prompt assembly and LVLM inference. This decoupling enables quick what-if analyses (e.g., adjusting the similarity threshold or per-modality token caps) without re-encoding frames. We persist per-video indices with time spans and modality tags and apply a deterministic round-robin budget across modalities to keep prompts stable across turns, which reduces run-to-run variance and simplifies deployment.

Prompt formatting and controls. Retrieved texts are linearized into short, time-stamped sentences grouped by modality. We order DET before ASR before OCR so the model establishes scene semantics before speech and fine-grained reading. To limit redundancy we deduplicate near-duplicate n-grams across modalities and drop overlapping time spans, then allocate a per-modality token cap with a round-robin budget to prevent any single source from dominating the context. The LVLM prompt includes a fixed header instructing that retrieved texts are auxiliary evidence rather than ground truth and to prefer visual cues when conflicts arise. In qualitative audits this reduced cross-talk between modalities and improved answer stability without introducing new heuristics at inference time.

Generality. Similar trends hold when swapping the retriever (e.g., cosine over CLIP text features) or varying the detection vocabulary; absolute scores change modestly, but the gap over no-retrieval baselines persists. This suggests the benefit stems from the retrieval-and-ordering mechanism rather than a particular backbone or index, and indicates architecture-agnostic utility. **Domain transfer and latency.** Without finetuning, the pipeline transfers across sports, vlogs, lectures, and cooking videos without per-domain knobs. Errors concentrate when music or crowd noise degrades ASR or when stylized fonts hurt OCR; in those cases DET anchors still provide useful cues. Indexing at ingestion time and reusing per-video stores keeps retrieval composition stable across queries for a given video, which simplifies interactive use. Contriever encoding for 1–2K snippets adds a small constant cost and FAISS range search is negligible relative to LVLM decoding; the dominant cost remains LVLM decoding. Batching videos amortizes indexing, and short, clause-like snippets reduce co-reference drift while improving human auditability.

6 Conclusion

We presented Video-RAG, a training-free, visually aligned retrieval augmentation that prepends query-relevant ASR/OCR/object texts alongside frames. By guiding cross-modal reasoning within fixed context budgets, Video-RAG yields consistent gains on Video-MME, MLVU, and LongVideoBench with modest token overhead, achieving 77.4% on Video-MME with LLaVA-Video-72B and surpassing GPT-4o on that benchmark. These results highlight the efficacy of targeted retrieval for long video understanding without additional training.

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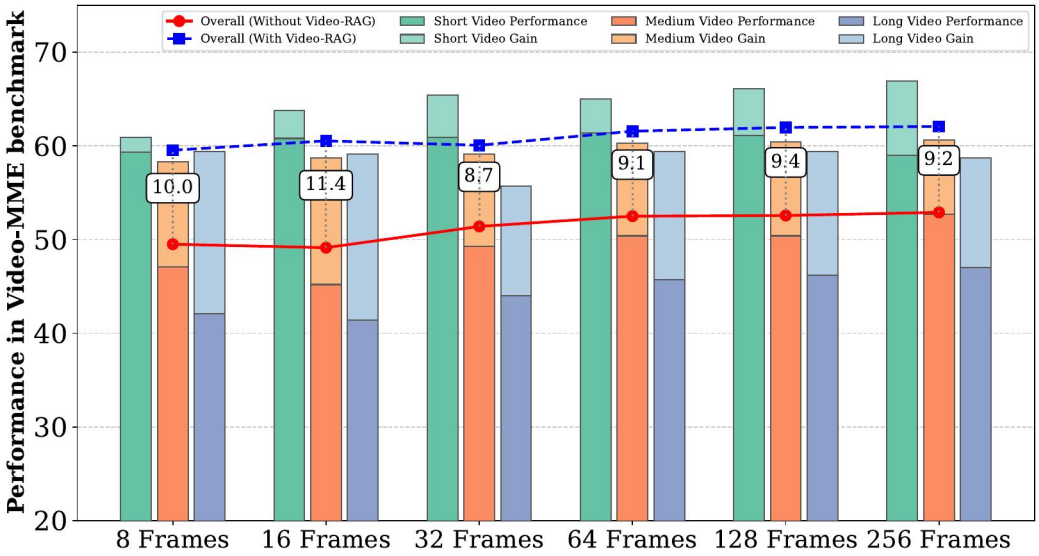

Figure 3: Effect of frame sampling rate on Video-MME with LongVA-7B [24]. Video-RAG complements frame selection by injecting query-aligned texts.

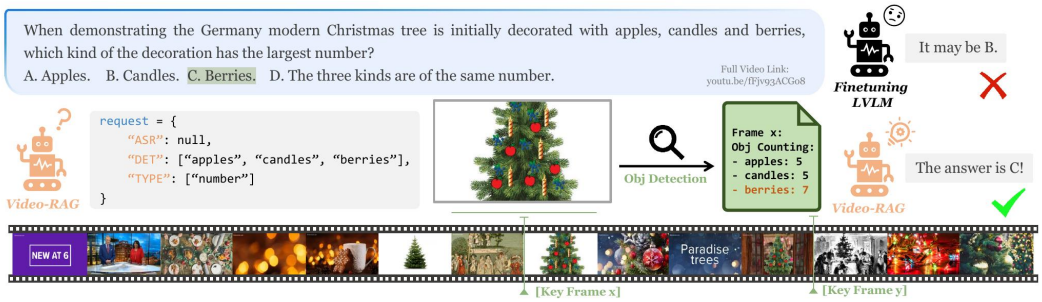

Figure 4: Qualitative example on Video-MME with LLaVA-Video [26]. Retrieved texts steer attention to query-relevant frames and entities.

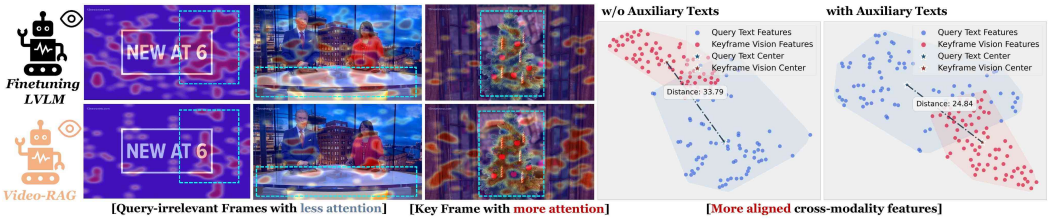

Figure 5: Grad-CAM heatmaps on the last hidden state and t-SNE of query/keyframe features for the example in Figure 4. Video-RAG sharpens alignment between the query and keyframes.

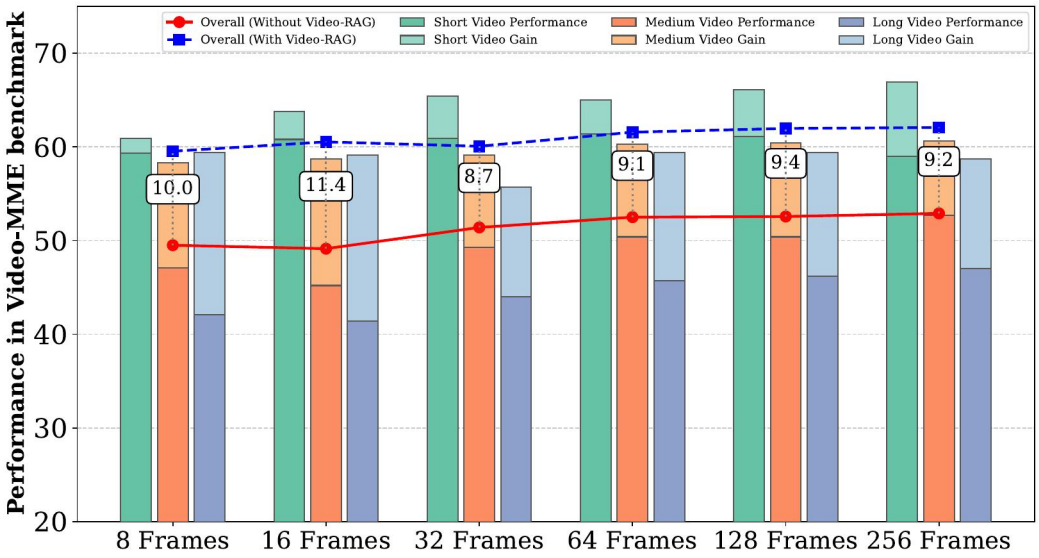

Figure 6: Frame-rate ablation on MLVU and LongVideoBench: Video-RAG favors sparser sampling by adding query-aligned texts and lowers decoding cost.

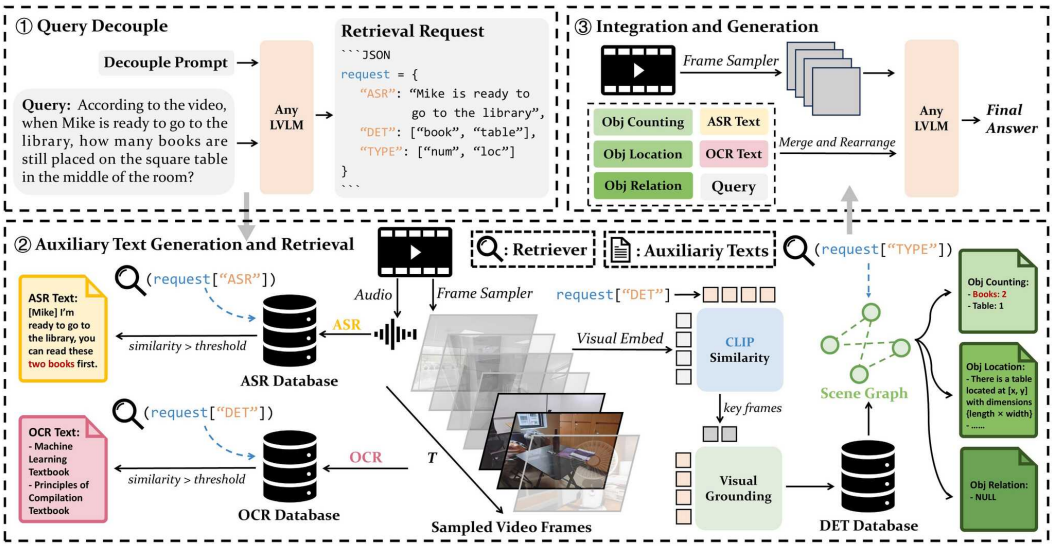

Figure 7: End-to-end pipeline with modality budgets and ordering: per-video stores (DET→ASR→OCR), FAISS index, Contriever retrieval, and round-robin caps.

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